Power Question 2009: Sign on the Label

An ***n*-label** is a permutation of the numbers 1 through n. For example, $J = 35214$ is a 5-label and $K = 132$ is a 3-label. For a fixed positive integer p, where $p \leq n$, consider consecutive blocks of p numbers in an n-label. For example, when $p = 3$ and $L = 263415$, the blocks are $263, 634, 341$, and 415. We can associate to each of these blocks a p-label that corresponds to the relative order of the numbers in that block. For $L = 263415$, we get the following:

$$\underline{263}415 \to 132; \quad 2\underline{634}15 \to 312; \quad 26\underline{341}5 \to 231; \quad 263\underline{415} \to 213.$$

Moving from left to right in the n-label, there are $n - p + 1$ such blocks, which means we obtain an $(n - p + 1)$-tuple of p-labels. For $L = 263415$, we get the 4-tuple $(132, 312, 231, 213)$. We will call this $(n - p + 1)$-tuple the ***p*-signature** of L (or **signature**, if p is clear from the context) and denote it by $S_p[L]$; the p-labels in the signature are called **windows**. For $L = 263415$, the windows are $132, 312, 231$, and 213, and we write

$$S_3[263415] = (132, 312, 231, 213).$$

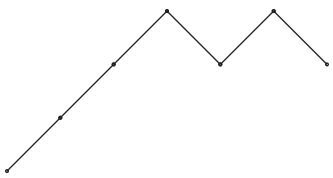
More generally, we will call *any* $(n - p + 1)$-tuple of p-labels a p-signature, even if we do not know of an n-label to which it corresponds (and even if no such label exists). A signature that occurs for exactly one n-label is called **unique**, and a signature that doesn't occur for any n-labels is called **impossible**. A **possible** signature is one that occurs for at least one n-label.
In this power question, you will be asked to analyze some of the properties of labels and signatures.

The Problems

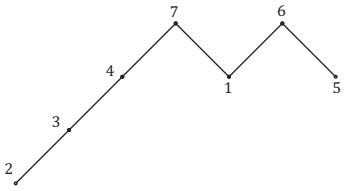
1. (a) Compute the 3-signature for 52341. [1]
   (b) Find another 5-label with the same 3-signature as in part (a). [2]
   (c) Compute two other 6-labels with the same 4-signature as 462135. [2]

2. (a) Explain why the label 1234 has a unique 3-signature. [1]
   (b) List three other 4-labels with unique 3-signatures. [1]
   (c) Explain why the 3-signature $(123, 321)$ is impossible. [1]
   (d) List three other impossible 3-signatures that have exactly two windows. [1]

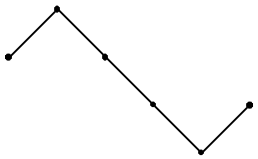
We can associate a **shape** to a given 2-signature: a diagram of up and down steps that indicates the relative order of adjacent numbers. For example, the following shape corresponds to the 2-signature $(12, 12, 12, 21, 12, 21)$:



A 7-label with this 2-signature corresponds to placing the numbers 1 through 7 at the nodes above so that numbers increase with each up step and decrease with each down step. The 7-label 2347165 is shown below:



3. Consider the shape below:



(a) Find the 2-signature that corresponds to this shape. [2]

(b) Compute two different 6-labels with the 2-signature you found in part (a). [3]

4. (a) List all 5-labels with 2-signature $(12, 12, 21, 21)$. [2]

(b) Find a formula for the number of $(2n + 1)$-labels with the 2-signature [3]

$$(\underbrace{12, 12, \ldots, 12}_{n}, \underbrace{21, 21, \ldots, 21}_{n}).$$

5. (a) Compute the number of 5-labels with 2-signature $(12, 21, 12, 21)$. [2]

(b) Determine the number of 9-labels with 2-signature

$$(12, 21, 12, 21, 12, 21, 12, 21).$$

Justify your answer. [3]

6. (a) Determine whether the following signatures are possible or impossible:

(i) $(123, 132, 213)$, [1]

(ii) $(321, 312, 213)$. [1]

(b) Notice that a $(p + 1)$-label has only two windows in its p-signature. For a given window ω_1, compute the number of windows ω_2 such that $S_p[L] = (\omega_1, \omega_2)$ for some $(p + 1)$-label L. [1]

(c) Justify your answer from part (b). [2]

7. (a) For a general n, determine the number of distinct possible p-signatures. [4]

   (b) If a randomly chosen p-signature is 575 times more likely of being impossible than possible, determine p and n. [1]

8. (a) Show that $(312, 231, 312, 132)$ is not a unique 3-signature. [1]

   (b) Show that $(231, 213, 123, 132)$ is a unique 3-signature. [2]

   (c) Find two 5-labels with unique 2-signatures. [1]

   (d) Find a 6-label with a unique 4-signature but which has the 3-signature from part (a). [2]

9. (a) For a general $n \geq 2$, compute all n-labels that have unique 2-signatures. [1]

   (b) Determine whether or not $S_5[495138627]$ is unique. [1]

   (c) Determine the smallest p for which the 20-label

   $$L = 3, 11, 8, 4, 17, 7, 15, 19, 6, 2, 14, 1, 10, 16, 5, 12, 20, 9, 13, 18$$

   has a unique p-signature. [3]

10. Show that for each $k \geq 2$, the number of unique 2^{k-1}-signatures on the set of 2^k-labels is at least 2^{2^k-3}. [5]